Exam Calculus A

202001171 (5 EC) / 202001172 (4 EC)

10 July 2025
8:45-11:45h

This exam consists of 9 exercises for 202001172 and of 11 exercises for 202001171.
Motivate all your answers.

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1. Find
$$\lim_{h\to 0^-} \frac{h}{\sin(3h)}$$

2. Find the derivative of f and express it in its most simple form.
$$f(x) = \frac{\sin x}{1-\cos x}$$

3. Find
$$\lim_{y\to 0} \frac{\sqrt{ay+a^2}-a}{y}, \quad a>0$$

4. Evaluate the following integrals

   (a)
   $$\int_{-\pi/2}^{0} \frac{\sin w}{(3+2\cos w)^2} dw$$

   (b)
   $$\int_0^1 \frac{dx}{\sqrt{1-x}}$$

5. Given are the following three lines in space:

   L1: $x = 1+2t, \quad y = -1-t, \quad z = 3t; \qquad -\infty < t < \infty$
   L2: $x = 2-s, \quad y = 3s, \quad z = 1+s; \qquad -\infty < s < \infty$
   L3: $x = 5+2r, \quad y = 1-r, \quad z = 8+3r; \quad -\infty < r < \infty$

   (a) Are L1 and L3 parallel, skew, or do they intersect? If they intersect, what is the intersection point?
   (b) Are L2 and L3 parallel, skew, or do they intersect? If they intersect, what is the intersection point?

6. Find the length of the curve $x = t^3, y = 1.5t^2, 0 \le t \le \sqrt{3}$.

7. $\mathbf{r}(t)$ is the position of a particle in space at time t.

$$\mathbf{r}(t) = (1+t)\mathbf{i} + \frac{t^2}{\sqrt{2}}\mathbf{j} + \frac{t^3}{3}\mathbf{k}$$

   (a) Find the particle's velocity vector.

   (b) Find the particle's acceleration vector.

   (c) What is the particle's speed at time $t = 1$?

   (d) What is the particle's direction of motion at time $t = 1$?

8. Given is $f(x, y, z) = x^2 + 2y^2 - 3z^2$; $P_0(1, 1, 1)$; $\mathbf{u} = \mathbf{i} + \mathbf{j} + \mathbf{k}$. Find the derivative of f at P_0 in the direction of $\mathbf{u}$.

9. Solve the differential equation

$$e^{2x}y' + 2e^{2x}y = 2x$$

Exercises 10 and 11 are only for 202001171.

10. Find all solutions of the equation $x^4 - 5x^2 + 25 = 0$. Give your answer in Cartesian form.

11. Find the unique solution of the following second-order initial value problem.

$$y'' + 12y = 0,\ y(0) = 0,\ y'(0) = 1$$

Norm:

Exercise	1	2	3	4a	4b	5a	5b	6	7a	7b	7c	7d	8	9	10	11	total
202001171	3	3	3	4	3	2	2	4	1	1	1	1	4	4	5	4	45
202001172	3	3	3	4	3	2	2	4	1	1	1	1	4	4	-	-	36

Exam grade 202001171: 1 + (obtained points)/5
Exam grade 202001172: 1 + (obtained points)/4